

# Submodules Catalogue — `vasic-digital` + `HelixDevelopment`

## Submodules Catalogue — vasic-digital + HelixDevelopment

| Field          | Value                                                                                                                                                                                                                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revision       | 1                                                                                                                                                                                                                                                                                                                                                                             |
| Created        | 2026-05-20                                                                                                                                                                                                                                                                                                                                                                    |
| Last modified  | 2026-05-20                                                                                                                                                                                                                                                                                                                                                                    |
| Status         | active                                                                                                                                                                                                                                                                                                                                                                        |
| Status summary | Initial catalogue. Enumerates every owned-by-us submodule under the two canonical organisations per Universal §11.4.74 (submodule-catalogue-first discovery). 142 repos total (122 + 20). Categorised by capability so consuming projects can locate "is there already a Submodule for X?" in one glance. Re-generate via <code>gh repo list</code> + the script in §3 below. |
| Issues         | none                                                                                                                                                                                                                                                                                                                                                                          |
| Issues summary | —                                                                                                                                                                                                                                                                                                                                                                             |
| Fixed          | initial catalogue creation                                                                                                                                                                                                                                                                                                                                                    |
| Fixed summary  | per user mandate 2026-05-20 follow-up to §11.4.76.                                                                                                                                                                                                                                                                                                                            |
| Continuation   | Re-enumerate quarterly (or after any non-trivial repo creation/ rename in either org).                                                                                                                                                                                                                                                                                        |

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## §1. Why this file exists

Per Universal §11.4.74 (Submodule-catalogue-first discovery + extend-don't-reimplement), every consuming project MUST survey these two organisations BEFORE scaffolding a new module. Without a single canonical inventory the survey becomes "open the GitHub page and skim 142 repos" — slow, easy to miss, error-prone.

This file is that inventory. It is the **first stop** for any "do we already have something that does X?" question. The full discipline is defined in Constitution.md §11.4.74; this file is the operational companion.

Per Universal §11.4.76, the vasic-digital/containers Submodule is the canonical container-orchestration substrate — every containerised workload across every consuming project goes through it.

## §2. How to consume the catalogue

When scaffolding a new feature in any consuming project:

1. Find the closest capability match in §4 or §5 below.
2. If a match exists, add the submodule to the consuming project at the parent-root path (<root>/<name>/ or <root>/submodules/<name>/) per Universal §11.4.28 (Submodules-As-Equal-Codebase).
3. Record the verdict in the tracker row: Catalogue-Check: reuse <org/repo>@<sha> or Catalogue-Check: extend <org/repo>@<sha>.
4. If extending: open a PR against the upstream Submodule (never duplicate in-project per §11.4.74).
5. If no match: record Catalogue-Check: no-match <YYYY-MM-DD> and justify in the PR description — operator-side validation re-checks the catalogue before approval.

The no-match path is rare — most "I need a small utility" needs are already covered by digital.vasic.commonshaped modules below.

## §3. Regeneration

Re-generate the lists in §4 / §5 by running:

```
gh repo list vasic-digital --limit 200 --json
    name,description,updatedAt > /tmp/v.json
gh repo list HelixDevelopment --limit 200 --json
    name,description,updatedAt > /tmp/h.json
```

then categorise via the python helper kept inline in this commit (scripts/build\_catalogue.py is the planned canonical tool — currently inline-built per the §11.4.74 launch commit). Update mtime in the metadata table on every regen.

## §4. vasic-digital catalogue (122 repos)

### Container + lifecycle

- **AgentWrapper** — Wrap AI CLI Coding Agents in Docker containers
- **Agentic** — Graph-based agentic workflow orchestration
- **AutoTemp** — SCAFFOLD / WIP -- Benchmark-driven temperature auto-tuning orchestration
- **BackgroundTasks** — Background Tasks module
- **HelixQA** — AI-driven QA orchestration for multi-platform testing
- **HyperTune** — SCAFFOLD / WIP -- Hyperparameter tuning orchestration
- **LLMOrchestrator** — Headless CLI agent management for LLM orchestration
- **tmux** — Optimized + verified containerized tmux build — reproducible across hosts, jemalloc-aware, OOM-protected, 8-test verification gate. Reusable on any Linux system.

### AI / LLM provider + agent

- **Android-Toolkit** — Commonly used set of abstractions and implementations.
- **Benchmark** — LLM benchmarking: SWE-bench, HumanEval, MMLU, leaderboard
- **Chutes-Toolkit** — Chutes Toolkit
- **Embeddings** — Generic reusable Go module: digital.vasic.embeddings
- **I-LLM** — SCAFFOLD / WIP -- Introspection layer for LLM providers
- **JVM-Toolkit** — Commonly used set of abstractions and implementations.
- **LLMGateway** — (no description)
- **LLMops** — LLM operations: evaluation, experiments, datasets, prompt versioning
- **LLMProvider** — Shared LLM provider interface, 40+ provider adapters, retry, circuit breaker, health monitoring
- **LLMsVerifier** — Benchmark and verify LLMs
- **Memory** — Generic reusable Go module: digital.vasic.memory
- **Normalize** — Adversarial-input canonicalisation library for defensive LLM guardrail pipelines
- **Open-Rag-Material-Android** — Material for RAG for Android development
- **Optimization** — Generic reusable Go module: digital.vasic.optimization
- **Planning** — AI planning algorithms: HiPlan, MCTS, Tree of Thoughts
- **RAG** — Generic reusable Go module: digital.vasic.rag
- **RedTeam** — YAML-driven adversarial prompt fixture harness for defensive LLM guardrail regression testing
- **SelfImprove** — AI self-improvement: reward modelling, RLHF, optimizer
- **SiliconFlow-Toolkit** — SiliconFlow Toolkit
- **SkillRegistry** — CLI agent skill registration and management for AI agent systems
- **Storage** — Generic reusable Go module: digital.vasic.storage
- **TOON** — Generic reusable Go module: digital.vasic.toon - Token-Oriented Object Notation wrapper
- **ToolSchema** — Generic tool schema definition, validation, and execution for AI agent tool systems
- **VectorDB** — Generic reusable Go module: digital.vasic.vectordb
- **VisionEngine** — Computer vision and LLM Vision for UI analysis and navigation

- **conversation** — Conversation context management, infinite context compression, and event sourcing for AI agents

## Messaging + observability + storage

- **Document** — Go document model with format detection and change tracking
- **EventBus** — Generic reusable Go module: digital.vasic.eventbus
- **Filesystem** — digital.vasic.filesystem - Reusable Go module
- **Formatters** — Generic reusable Go module: digital.vasic.formatters
- **I18n** — digital.vasic.i18n - Generic internationalization Go module
- **Lazy** — Generic reusable Go module: digital.vasic.lazy - Lazy loading with sync.Once generics
- **MCP\_Module** — Generic reusable Go module: digital.vasic.mcp
- **Messaging** — Generic reusable Go module: digital.vasic.messaging
- **Plugins** — Generic reusable Go module: digital.vasic.plugins
- **Streaming** — Generic reusable Go module: digital.vasic.streaming
- **cache** — Generic reusable Go module: digital.vasic.cache
- **concurrency** — Generic reusable Go module: digital.vasic.concurrency
- **config** — digital.vasic.config - Reusable Go module
- **database** — Generic reusable Go module: digital.vasic.database
- **discovery** — digital.vasic.discovery - Reusable Go module
- **http3** — Generic Go module wrapping quic-go/http3 for net/http.Handler servers — drop-in HTTP/3 support
- **mdns** — Generic Go module for RFC 6762/6763 mDNS service announcement and discovery — drop-in LAN service registration
- **observability** — Generic reusable Go module: digital.vasic.observability
- **ratelimiter** — digital.vasic.ratelimiter - Reusable Go module
- **recovery** — Generic reusable Go module: digital.vasic.recovery - Application-level fault tolerance
- **tracker\_sdk** — Generic, tracker-agnostic SDK primitives: mirror manager, plugin registry, testing harness. Used by the Lava project (<https://github.com/milos85vasic/Lava>) to support multiple torrent trackers.

## Auth + security + middleware

- **Claritas** — SCAFFOLD / WIP -- System-prompt extraction detection
- **GandalfSolutions** — SCAFFOLD / WIP -- Read-only solutions archive for prompt-leak-defense testing
- **LeakHub** — SCAFFOLD / WIP -- Prompt-leak corpus / defensive-use fixtures (red-team training)
- **Ouroborous** — SCAFFOLD / WIP -- Recursive/self-referential safety patterns
- **Veritas** — SCAFFOLD / WIP -- Truth/verification auxiliary
- **auth** — Generic reusable Go module: digital.vasic.auth
- **middleware** — digital.vasic.middleware - Reusable Go module
- **security** — Generic reusable Go module: digital.vasic.security

## Cross-platform (KMP) modules

- **Auth-KMP** — Kotlin Multiplatform OAuth2 authentication: flows, token management, secure storage interface
- **Concurrency-KMP** — Kotlin Multiplatform concurrency utilities: lazy loading, platform synchronization, flow-based loaders

- **Config-KMP** — Kotlin Multiplatform storage configuration types for network protocols
- **Database-KMP** — digital.vasic.database - KMP network storage database interfaces and entity types
- **Document-KMP** — Kotlin Multiplatform document model with format detection and change tracking
- **Formatters-KMP** — Cross-platform KMP text format detection, parsing interfaces, and format registry
- **RateLimiter-KMP** — Kotlin Multiplatform rate limiting: semaphore, token bucket, adaptive, throttler
- **Security-KMP** — Kotlin Multiplatform secure storage: AES encryption, platform Keychain/KeyStore integration
- **Storage-KMP** — digital.vasic.storage - KMP network storage service interfaces and abstractions
- **UI-Components-KMP** — Kotlin Multiplatform UI components: theme system, animations, accessibility utilities for Compose

## TypeScript / React + Web

- **API-Client-TS** — Generic REST API client for TypeScript
- **Auth-Context-React** — React AuthProvider and useAuth hook for Catalogizer authentication with React Query
- **Calling-Engine-TS** — Generic SIP.js + WebRTC calling engine
- **Catalogizer-API-Client-TS** — Type-safe TypeScript client for the Catalogizer API with axios, auth, retry, and per-domain services
- **Collection-Manager-React** — React collection management components for Catalogizer
- **Dashboard-Analytics-React** — React dashboard and analytics components for Catalogizer
- **I18n-Client-TS** — Generic client-side i18n wrapper around i18next
- **Media-Browser-React** — React entity browser components for Catalogizer media browsing
- **Media-Player-React** — React media player component for Catalogizer entity playback
- **Media-Types-TS** — Shared TypeScript type definitions for Catalogizer media entities, auth, and API
- **State-Management-TS** — Generic Redux Toolkit state management utilities
- **Testing-Utils-TS** — Generic React + Redux testing utilities
- **UI-Components-React** — Reusable TypeScript/React module
- **WebSocket-Client-TS** — Reusable TypeScript/React module

## Testing + QA + benchmarking

- **Panoptic** — A comprehensive tool for automated testing, UI recording, and screenshot capture across web, desktop, and mobile applications.
- **ReplayBuffer** — Reusable ReplayBuffer module for visual testing and automation
- **ScreenDiff** — Reusable ScreenDiff module for visual testing and automation
- **TrainingCollector** — Reusable TrainingCollector module for visual testing and automation
- **VisualRegression** — Reusable VisualRegression module for visual testing and automation
- **challenges** — (no description)

## Apps / tools / experimental

- **ATMOSphere-Android-15** — (no description)
- **ATMOSphere-Android-15-Main** — Android 15 AOSP for Orange Pi 5 Max
- **Asinka** — Interprocess Objects Synchronization Library: Asinka (Асинка)
- **Assets** — Generic, reusable Go module for lazy asset loading with strategy-based resolution
- **Catalogizer** — Advanced Multi-Protocol Media Collection Management System
- **Courses-Creator** — Courses creator toolkit
- **DocProcessor** — Documentation processing and feature map extraction for QA automation
- **Entities** — Generic media entity system: title parsing, media types, hierarchy detection, and aggregation primitives for Go
- **Games** — Vasic Digital Games
- **GrabTube** — Tube services downloader
- **HelixAgent** — (no description)
- **Integrations** — Integrations
- **Media** — digital.vasic.media - Reusable Go module
- **Models** — (no description)
- **Network-Binder** — Bind multiple internet connection endpoints into single one using mptcp.
- **OOM-Protect** — Workstation OOM hardening for systemd Linux
- **PliniusCommon** — SCAFFOLD / WIP -- Common types/errors library (shared foundation for the 8 sibling modules)
- **Proxy** — Proxy server
- **Pure\_v2\_Firmware** — Firmware for BeagleBone
- **SDK** — SDK
- **ShareConnect** — Share the downloadable URLs to remote (or local) processing endpoints
- **TransmissionConnect** — Remote control for Transmission BitTorrent client and integration with ShareConnect
- **Watcher** — digital.vasic.watcher - Reusable Go module
- **Yole** — Text editor, Notes & ToDo - Markdown, todo.txt, plaintext, math, and much more ...
- **ffmpeg-kit** — FFmpeg Kit for applications. Supports Android, Flutter, iOS, Linux, macOS, React Native and tvOS. Supersedes MobileFFmpeg, flutter\_ffmpeg and react-native-ffmpeg.
- **qBitConnect** — Control qBittorrent from any device and integrate with ShareConnect

## Misc / scaffolds

- **Herald** — Ingesting system events and reliably fanning them out to multiple notification channels so every alert reaches the right destination without confusion
- **containers** — (no description)
- **vasic-digital.github.io** — (no description)

## §5. HelixDevelopment catalogue (20 repos)

- **DebateOrchestrator** — Multi-agent debate orchestration library — Phase 1 reconstruction (7 real + 5 stub sub-packages; see RECONSTRUCTION\_ROADMAP.md)
- **DocProcessor** — Documentation processing and feature map extraction for QA automation
- **Helix-Game-1** — First Stake game from HelixDevelopment
- **Helix-Game-Demo** — (no description)
- **HelixAgent** — LLMs Agent
- **HelixBuilder** — AI powered application building pipeline
- **HelixCode** — AI Coding Agent
- **HelixConstitution** — Constitution, AGENTS.MD and CLAUDE.MD universal agent rules and constraints
- **HelixGitpx** — Helix Git Proxy eXtended
- **HelixLLM** — Helix LLM - Local running super model
- **HelixMemory** — Super Memory Provider — Unified Cognitive Memory Engine fusing Mem0 + Cognee + Letta for HelixAgent
- **HelixPlay** — Ultimate gaming experience!
- **HelixSpecifier** — Spec-Driven Development Fusion Engine - Unified specification intelligence fusing SpecKit + Superpowers + GSD for HelixAgent AI debate ensemble
- **HelixTranslate** — A high-performance, enterprise-grade universal ebook translation toolkit supporting any ebook format and any language pair, featuring multiple translation engines, REST API with HTTP/3 support, and real-time WebSocket events.
- **LLMOrchestrator** — Headless CLI agent management for LLM orchestration
- **LLMProvider** — Shared LLM provider interface, 40+ provider adapters
- **Projects** — Helix projects
- **Stake-Tetris** — Stake Tetris
- **VisionEngine** — Computer vision and LLM Vision for UI analysis and navigation
- **helixqa** — AI-driven QA orchestration for multi-platform testing